

CAP777:WEB DEVELOPMENT USING PHP

L:3 T:0 P:2 Credits:4

Course Outcomes: Through this course students should be able to

- CO1 :: knowledge about dynamic web-based solutions development
- CO2 :: understand essentials skills for commercial web development
- CO3 :: Apply PHP and MySQLi to develop dynamic web pages by implementing functions
- CO4 :: Analyze database-driven web applications by integrating PHP with MySQLi
- CO5 :: Create complete and secure web applications using Laravel with advanced features

Unit I

Basics of PHP : introduction to PHP, client site script vs server site script, compiling process for php web page, variables in php, constants in php, control flow statements, looping statements, OOPs in PHP

Unit II

Functions in PHP : built in functions in php, defining and calling user defined functions in php, function parameters in php

Unit III

Arrays in PHP : types of arrays supported in php, performing operations on array in php, traversing arrays in php, sorting arrays in php, array functions in php

Strings in PHP : single-quote strings, double-quote strings, built-in String functions

Unit IV

Handling HTML forms with PHP : html form controls, capturing HTML form data using PHP, sending data from PHP to HTML page

Unit V

Database using MySQLi with PHP : database options using php, performing database operations using mysqli, performing database navigation using mysqli, using grid with mysqli

Unit VI

Introduction to Laravel Framework : configuring laravel framework, html template to laravel blade template, database migrations

Working with Laravel framework : migrating from HTML to Laravel, managing Laravel versions using Git Hub

Working with Graphics and PDF : implementing graphics and pdf in php, creating and drawing images, inserting images and graphics in pdf

List of Practicals / Experiments:

Pracitcals

- Implement a PHP program that accepts user input from an HTML form and processes it to display a formatted message based on the input. Ensure proper validation and error handling
- Write a PHP script that demonstrates the use of both indexed and associative arrays. Include operations such as adding, removing, and updating elements, and display the results in an HTML table.
- Develop a PHP class that models a real-world entity (e.g., a library book or a user profile). Include methods for setting and retrieving properties, and demonstrate inheritance by creating a subclass with additional functionality.
- Design a PHP application that connects to a MySQLi database, retrieves user data, and displays it on a webpage. Include functionality to search, filter, and sort the data using SQL queries.
- Build a simple CRUD (Create, Read, Update, Delete) application in PHP that interacts with a MySQLi database. Implement features to add new records, edit existing ones, and delete records from the database.
- Implement a PHP script that demonstrates the use of sessions and cookies to manage user authentication. Create a login system that maintains user state across different pages using sessions.

- Write a PHP function that generates a random password of a specified length. The password should include a mix of uppercase letters, lowercase letters, numbers, and special characters. Provide examples of the function's output and discuss how to ensure the generated passwords are secure
- Write a PHP script to accept a string from an HTML form, convert it to uppercase, replace all spaces with underscores, and count the number of characters in the modified string.

Text Books:

1. PROGRAMMING PHP by RASMUS LERDORF, KEVIN TATROE, O'REILLY

References:

1. WEB TECHNOLOGIES BLACK BOOK by KOGENT LEARNING SOLUTIONS INC., DREAMTECH PRESS
2. WEB TECHNOLOGIES HTML, CSS3, JAVASCRIPT, JQUERY, AJAX, PHP, XML, MVC AND LARAVEL by DT EDITORIAL SERVICES, DREAMTECH PRESS